How Laya Chooses

2026-10-09 · Vignesh

Laya's options do compete, but order flips come from positional encoding, not competition; confidence is set by three late MLPs; and noul answers follow semantic label words because the [MASK] reads its label in local layers before passage evidence arrives.

At a glance

C3 and C4 held outright, C1 and C2 held in a narrower form than stated, C5 held partly; the one clean null is a confidence *direction* in the residual stream. All headline numbers replicate in fp32 and on held-out data.

Claim	Verdict	Key number
C1 Options compete	Yes, distributed	Mean change of another option's logit when one option is blanked: 0.37
C1 Order flips come from competition	No: from positions	Flips stay at 28.3% with options blocked from each other; 0.0% when blocked options also share positions
C1-Control (parallel layout)	Confirms the split	Blocking option attention halves competition, leaves flips
C2 Choice and confidence separable	At component level	Ablating MLPs L22, L26, L27: confidence ×0.48, 96% of answers kept
C2 at direction or feature level	No	Steering a confidence direction = random; SAE features entangled
C3 Label vs state competition	Yes, semantic	yes/no labels swapped → 96% follow the label; A/B → 3%
C4 Jailbreak: decodable early, decided late	Yes	Probe 92% at layer 0; decision fixed after layer 15–28
C5 Objective matters	Partly	3 seeds: RLCD 16% less competition (p = 0.09, a trend); architecture effects identical

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What we did: model, methods and timeline

All experiments ran on Modal (H100 GPUs for model work, CPU containers for setup) between 2026-10-08 and 2026-10-09, following the frozen study plan (v3.1, five claims, $100 cap).

The model under study

- **laya** (convaiinnovations/laya, package laya 0.4.1): a ModernBERT-large encoder (28 layers, width 1024, 16 attention heads; global attention on layers 0, 3, 6, …, 27 and a local ±64-token window on the others), followed by 2 standard transformer layers and an option scorer. 421M parameters.
- **Input layout:** `[CLS] <type> question: <instructions> [SEP] [MASK] option0 [MASK] option1 … [SEP] state [SEP]`, up to 512 tokens. Each option is scored at its own [MASK]; scores are divided by a per-(question type, option count) temperature and softmaxed.
- **Question types:** choice (pick one), noul (yes/no; options render as "false: no, the statement does not hold" and "true: yes, the statement holds"), score (ordinal).
- **Comparison models:** smolaya (20-layer variant trained with cross-entropy plus distillation and random option order) and laya-multilingual (mmBERT-base, 22 layers).

Step 1 · A verified instrument

Laya's code does not expose its internals, so its forward pass was rewritten with named points that can be recorded, replaced or blocked: the residual stream entering every layer, each head's output, each attention pattern, each MLP output, and custom attention masks per layer or per head.

Check (200 inputs)	Result
Raw logits vs laya's own model (eager attention)	max difference 3.8×10^{-6}
Probabilities vs laya's `predict` (rounded to 4 dp)	max difference 5×10^{-5} on laya, smolaya and laya-multilingual
Local attention window	±64 confirmed (±63 or ±65 give 0.2 logit errors)
Package's own "parallel" option layout	matched to 3×10^{-6}
bf16 vs fp32 answers	99–100% agreement; headline numbers re-run in fp32
Throughput, H100, 512 tokens, bf16	494 sequences/s

Step 2 · Data

Set	Source	Size (main / held-out)	Used for
Multiple choice	ARC-Challenge 500, ARC-Easy 300 (test), CommonsenseQA 689 (validation), QASC 500 (validation, cut to 4 options)	1,989 (1,392 / 597)	C1, C2, C5
Yes/no passages	BoolQ validation, passages ≤150 tokens	1,000 (700 / 300)	C3
GPT-written yes/no items	gpt-5.6-luna, answers checked by gpt-5.6-terra	971 (679 / 292)	C3 replication
Natural jailbreak set	in-the-wild prompts, jackhhao, ToxicChat, Alpaca	3,000 (2,100 / 900)	C4
Jailbreak minimal pairs	6 hand-written template families around benign requests; family 6 held out	1,016 kept of 1,700	C4 patching
Prompt injection	deepset/prompt-injections	662 (546 / 116)	C4
Rewritten jailbreaks	gpt-5.6-terra removes only the override	145	C4
Hindi pairs	NLLB-200 translation of 200 kept pairs	200 + 200	C4 multilingual
Retrain data	ARC train+validation, CSQA train, QASC train, BoolQ train	31,419	C5 controls

Multiple-choice items are cast as laya choice questions with the instruction "Which option correctly answers the question in the text?" and the question as the state; this casting scored best of three on a 300-item probe (0.46 vs 0.43 and 0.41).

Step 3 · Experiments, in the order they were run

Track	What was done
1 · Order and competition	Every item scored under all cyclic shifts, the reversal and 6 random orders. Competition matrix M by replacing one option's tokens with length-matched filler (two fillers tested) and by removing it. Parallel-layout controls and masking by layer band.
1b · Where and which heads	Attention summaries for every head; per-head screens (block cross-option attention, order-blind positions, mean ablation) over all 480 heads; information-flow patching layer by layer; causal tests on the top 5/10/20 heads against 3 random-head sets each.
2 · Choice vs confidence	Direct and tuned logit lenses; mean-ablation screen of all heads and MLPs; component sets for choice (A) and confidence (B), later chosen greedily; a confidence direction tested by steering; calibration and temperature analysis.
3 · Label-following	Label words swapped under four label types and both slot orders; strength and distance sweeps; patching of label vs state information; attention-edge knockouts in local, global and head layers.

Track	What was done
4 · Jailbreak and injection	Probes per layer with shuffled-label controls; segment patching on minimal pairs; Track 1–2 components ablated on jailbreak items; diff-of-means steering and ablation; Hindi transfer on laya-multilingual.
Follow-ups	fp32 confirmation of headline numbers; held-out replication of Track 3; greedy component search; steering sweep over layers and token positions; GPT-generated data.
Stretch	Two controls from the same ModernBERT-large start, same data, fixed option order, differing only in loss (cross-entropy vs RLCD), 3 seeds each; a TopK sparse autoencoder (16,384 latents) on 201k [MASK] vectors.
Fix	`label_guard`: the C3 edge block installed into stock laya and evaluated through its official `predict` API, with side-effect checks.
Literature check	Web search for the closest prior work on each claim (Related work section).

Errors found and corrected along the way

- The first confidence-steering test used doses in units of a tiny standard deviation and showed no effect; it was rescaled to the residual-stream norm and re-run (still a null).
- A loader bug crashed the first fp32 confirmation; it was fixed and re-run.
- A Windows text-encoding bug corrupted a few source lines; they were repaired before further runs.
- The "RLCD gives 20% less competition" figure came from one seed; with 3 seeds it is 16% ($p = 0.09$), reported as a trend.
- C1 wording: the 0.0% flip condition combines shared positions with blocked option attention; shared positions alone were not cleanly tested.
- C4 wording: steering is not "only late"; subtraction worked best at layer 12, addition at layer 20.
- SAE feature labels were contaminated by the labelling prompt and are not used as evidence.

C1 · Option competition vs order

Competition is real but does not cause order flips. Flips come from where an option sits (RoPE positions); competition flows option-text → option-text, then into the [MASK].

Setup: 2,000 multiple-choice items (ARC, CommonsenseQA, QASC, 3–5 options), cast as laya `choice` questions. Competition M = change in option i's centred logit when option j's tokens are replaced by length-matched filler.

Condition (laya, fp32)	Accuracy	Reversal flips	Mean abs(M)
Trained (sequential)	0.453	29.6%	0.374
Options cannot see each other	0.414	28.3%	0.181
Options blocked + shared positions (shipped parallel)	0.408	0.0%	0.175
Fully masked parallel	0.346	28.6%	0.000

- **Not a slot bias.** Slot index explains 0.2% of order-induced variance; neighbour strength correlates at $r \approx 0$.
- **Route.** Patching shows option j's content leaving its own span by layer 12, peaking in option i's text tokens at layer 16 (recovery 0.53), and reaching [MASK]i at layers 17–22 (0.28 → 0.84). The two head layers add nothing.
- **No small circuit.** Blocking cross-option attention in the top 5, 10 or 20 heads moves M from 0.371 only to 0.36, no more than random heads. The top competition heads (L0 h5–7, L4 h6–7) are shared with smolaya, likely from the ModernBERT init.
- **smolaya** shows the same split at half strength: M 0.194, flips 17.7% → 1.0% with options blocked and positions shared. Shared positions were not tested without the attention block; an order-blind-RoPE-only run cut flips to 8% but collapsed accuracy to chance, so it is not used as evidence.

C2 · Choice vs confidence

Three late MLPs act as a confidence amplifier: removing them halves the logit scale and fixes most of the miscalibration without changing answers. No compact set controls the choice itself.

Choice = direction of the centred logit vector; confidence = its length. Components were mean-ablated one at a time, then chosen greedily.

Ablation (laya, fp32)	Confidence scale	Answers kept	Calibration error (ECE)	AUROC of margin
None	×1.00	100%	0.19	0.63
MLPs L22, L26, L27	×0.48	95.7% (held-out 96.6%)	0.076	0.63
10 components, first pass	×0.29	91.5%	0.061	0.64
Greedy "choice" heads	≈×1.0	90.6%	0.20	0.63

- **Information vs scale.** AUROC is unchanged, so the amplifier adds sharpness, not knowledge of which answer is right.
- **Timing.** The choice direction stabilises by layer 22; in the direct lens the norm is only half its final size at layer 25 and jumps in the head layers.
- **Overconfidence is input-dependent.** The best temperature rises from 1.8 to 3.0 across logit-norm quintiles and is 4.8 on ARC-Challenge vs 2.0 on ARC-Easy, so one global temperature cannot fix it.
- **Nulls.** A residual direction that predicts confidence ($r \approx 0.7$) changes it no more than a random direction when steered. The 3-MLP set does not shrink jailbreak confidence (ratio 1.03); only the 10-component set did (0.36).

C3 · Label-following

When the label word disagrees with the option it sits on, laya follows the word, and only for meaningful words. The [MASK] reads its label in local layers 10–14; passage evidence arrives at layers 21–23. Cutting one attention edge removes the failure.

Test: noul questions where each option's label word is swapped ("true: no, the statement does not hold"). A model reading the passage is unaffected; a model reading the label word picks the option the passage contradicts.

Labels (BoolQ, 700 items, fp32)	Accuracy normal → swapped	Follows the label
yes / no	0.759 → 0.246	96%
true / false	0.756 → 0.581	39%
A / B	0.771 → 0.763	3%
◇ / △	0.770 → 0.764	4%

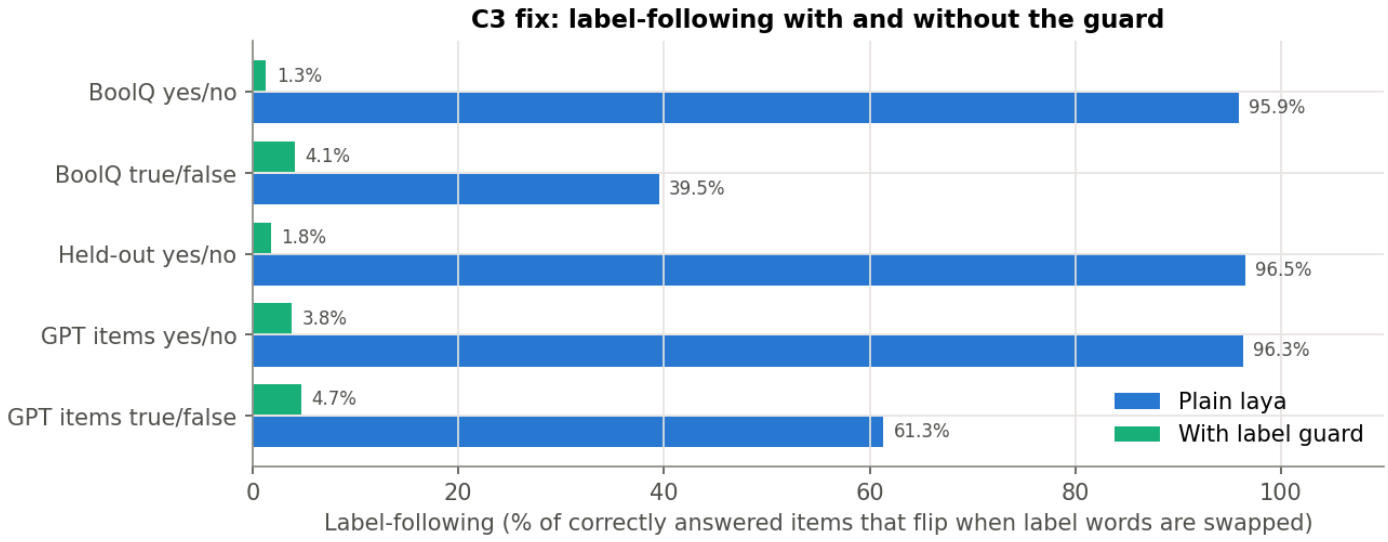
- **Replicated** on 300 held-out BoolQ items (42% for true/false) and on ~680 fresh items written by gpt-5.6-luna and verified by gpt-5.6-terra (96% yes/no, 60% true/false, 1–3% symbols).
- **Distance.** Pushing the passage 256 tokens further from the [MASK]s raises true/false label-following from 39% to 77%; A/B stays flat.
- **Fix.** Blocking [MASK] → own-label attention in local layers only drops label-following to 4% (held-out 3%, LLM items 4%) with normal accuracy unchanged or higher (0.785 → 0.811 on LLM items). The same block in global or head layers does nothing.

The fix as a drop-in patch

Installed on a stock `laya.load` agent (default SDPA attention, bf16), the label guard removes label-following through the official `predict` API. It blocks noul [MASK]s from their label words in local layers only.

Data	Label-following, plain → guard	Accuracy, normal labels, plain → guard
BoolQ yes/no	95.9% → 1.3%	0.759 → 0.756
BoolQ true/false	39.5% → 4.1%	0.753 → 0.759
Held-out BoolQ yes/no	96.5% → 1.8%	0.760 → 0.753
GPT-written items yes/no	96.3% → 3.8%	0.782 → 0.803
GPT-written items true/false	61.3% → 4.7%	0.783 → 0.812

Costs: choice questions are bit-identical; jailbreak AUROC falls 0.940 → 0.919 (1.8% of decisions change); injection AUROC 0.898 → 0.870 (no decisions change); 4 of 80 preset-style noul decisions change. Code: `interp/label_guard.py`.



C4 · Jailbreak, injection and Hindi

As in LLMs, a jailbreak is decodable from layer 0 but decided late. In laya the decision travels prompt → question tokens → [MASK], and it keys on jailbreak style more than on the override instruction itself.

- **Baselines.** Jailbreak AUROC 0.94 on 900 held-out natural prompts; injection AUROC 0.87 (deepset).
- **Decodable vs decisive.** A linear probe reads the label at 92% from layer 0 (shuffled-label control near chance). Patching jailbreak-run activations into the control run recovers the decision from prompt tokens only through layer 15; question tokens relay it at layers 14–20 (recovery up to 0.79); the [MASK] holds only about 0.4 until the first head layer completes it.
- **Steering depends on the layer.** Adding the jailbreak direction at the layer-20 [MASK]s raises benign prompts flagged from 23% to 61%, with an unrelated urgency question at 3% and multiple-choice accuracy unchanged. Subtracting it at the layer-12 prompt tokens lowers flagged jailbreaks from 98.5% to 72%. At layer 8 neither direction had a useful effect.
- **Style over instruction.** gpt-5.6-terra rewrote 145 flagged jailbreaks to remove only the override; laya still flags 82% of them (log-odds drop 7.7 but stay positive).
- **Shared signal.** Jailbreak and injection directions have cosine 0.71.
- **Hindi.** laya-multilingual flags 52% of Hindi-translated jailbreaks vs 8.5% of controls. The English-fit direction barely reads Hindi (AUROC 0.56) or steers it, but projecting it out drops Hindi detection from 52% to 8.5%: necessary, not sufficient.

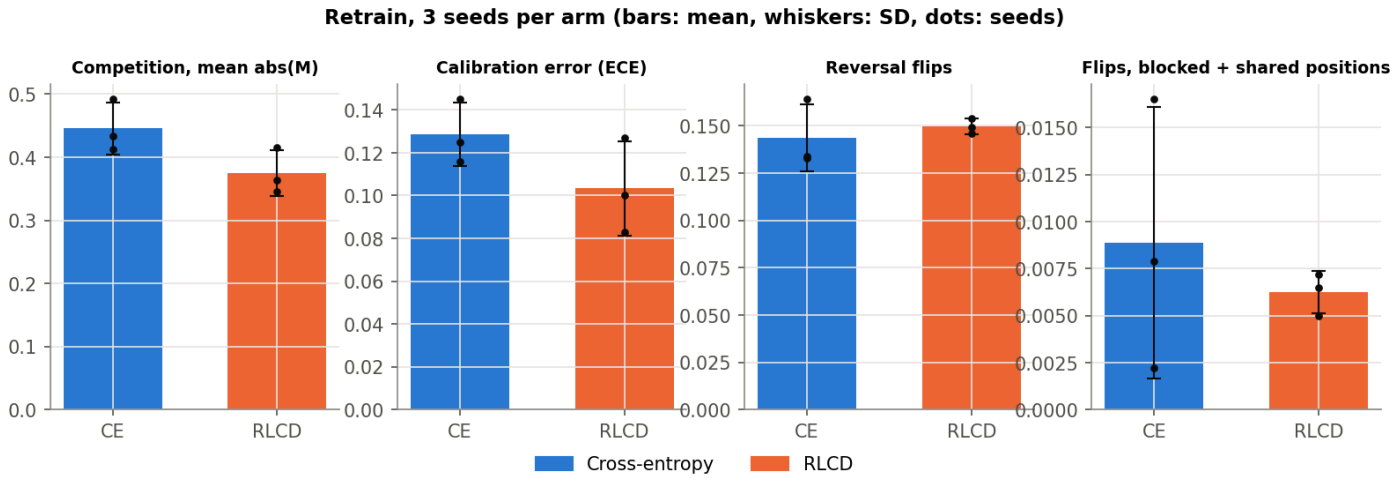
C5 · Training objective and SAE features

The position-driven order effect and the late-MLP confidence amplifier appear in fresh models trained with either objective, so they come from the architecture and layout. Laya's label-following does not: neither control shows it, so it comes from laya's training data.

Two controls, same ModernBERT-large init, same 31,419 training items (ARC, CSQA, QASC and BoolQ train splits), fixed option order, 2 epochs; only the loss differs. Each arm was trained with 3 seeds; the table shows seed 0.

Model	Accuracy	Reversal flips	Flips, shared positions	ECE	Mean abs(M)	Follows yes/no label
Control, cross-entropy	0.713	13.3%	0.8%	0.125	0.433	5%
Control, RLCD (laya's loss)	0.690	14.9%	0.7%	0.100	0.345	0.3%
laya	0.456	29.6%	0.0%	0.192	0.374	96%
smolaya	0.548	17.7%	1.0%	0.057	0.194	not run

- **Objective.** Over 3 seeds, RLCD has 16% less competition than CE (mean abs(M) 0.375 ± 0.036 vs 0.446 ± 0.041, Welch p = 0.09) and lower calibration error (0.103 vs 0.129, p = 0.19). Both are trends, not significant. Accuracy (0.695 vs 0.709), reversal flips (15.0% vs 14.4%) and label-following (4% vs 4%) do not differ.
- **Amplifier.** Greedy search finds a late-MLP confidence set in all 6 control models; every set contains layer 22 and the first head layer's MLP (index 28). Across all 6, blocking option attention leaves 15–17% flips and adding shared positions brings them under 2%.
- **SAE.** A TopK SAE (16,384 latents, k = 32) on 201k [MASK] vectors keeps 95% (layer 21) and 99% (layer 28) of answers when it replaces the real activations. The top-25 choice and confidence features overlap 11–15 of 25, and ablating the "choice" features also shrinks confidence. Choice and confidence are not separate features; they separate only at the component level.



Related work

The ideas exist for decoder LLMs; what is new here is the encoder decision-model setting, the C1 contrast, the C3 timing race and fix, and the controlled retrain.

Our result	Closest prior work	Difference
C1 order flips are positional	Pezeshkpour & Hruschka 2023; Zheng et al. 2023; biased heads in GPT-2 (Anchored Answers); Set-Based Prompting shares positions and masks attention	In their LLM ablations the attention mask removed more order dependency than positions; in laya blocking attention alone leaves flips at 28%
C2 confidence amplifier	Confidence Regulation Neurons (entropy neurons in LLMs)	Same idea in an encoder decision head, localized to late MLPs, present under both training losses

Our result	Closest prior work	Difference
C3 label-following	Wei et al. 2023 (semantic priors on labels); Label Words are Anchors	Adds the label-vs-evidence timing race, a one-edge fix, and evidence that it comes from training data
C4 decodable early, decided late	Wen et al. 2026, arXiv 2609.37737	Replicated in an encoder classifier; decision relayed through question tokens

These were found by web search on 2026-10-09; the papers were not all read in full.

Caveats and deviations from the plan

- **Off-task casting.** Laya was not built for multiple-choice QA (45% accuracy); C1 and C2 describe how it handles options, not a tuned QA behaviour.
- **Off-distribution controls.** Masked-parallel conditions lower accuracy (0.35–0.41), so they bound dependence rather than cleanly ablate it.
- **C3 data.** The plan's LLM-generated items were added after BoolQ, once the OpenAI key worked; both agree.
- **Rewritten jailbreaks** were not checked by hand; some may still contain jailbreak framing.
- **SAE feature labels** are contaminated (the labelling prompt showed whether each example won) and are not used as evidence. Its variance-explained figure is inflated by outlier dimensions; the answer-preservation check is the one that counts.
- **Retrain** uses 3 seeds per arm, which is enough for the architecture findings but leaves the objective effects at trend level; it uses public data, not laya's training set.
- **Not done:** "natural counterfactual" segment patching (the rewrites change token length), and the plan's browser-only workflow (runs were launched from the laptop CLI).

Cost, data and reproduction

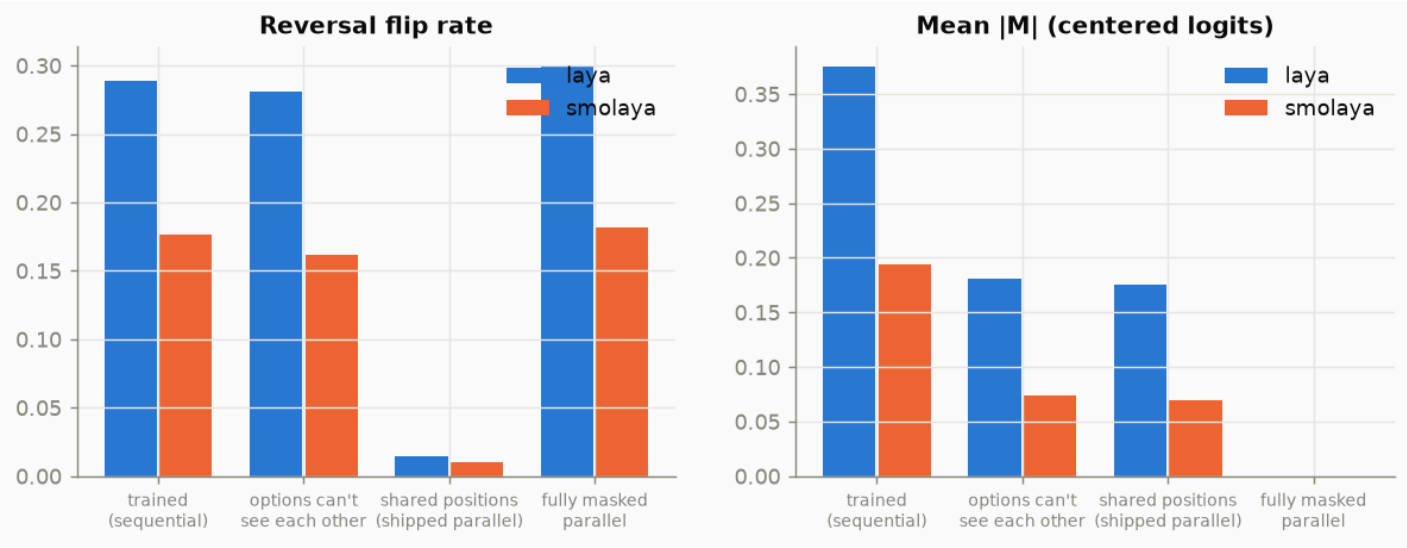
Total compute was $12.23 on Modal (all covered by free credits, $0 billed) against a $100 cap; OpenAI usage was about 160k prompt and 186k completion tokens.

- **Verified instrument.** An explicit forward pass of laya's DecisionModel matches `predict` to 5e-5 on laya, smolaya and laya-multilingual; local attention window ±64 confirmed.
- **Code.** `modal_interp.py` and the `interp/` package in the project repo (commits 1a02a62, f303893, 0b7c332).
- **Results** live on the Modal volume `laya-interp` under `results/`; the full figure report is served by the token-gated `laya-interp` web endpoint.
- **Rerun** any track with `modal run modal_interp.py --task <track1|track1b|track2|track3|track4|track43|followups|fp32|llm_steps|stretch|report>`.

Appendix · Figures and full tables from the Modal report

Generated on Modal from the stored result files (bf16 runs unless marked fp32). Tables show bootstrap 95% intervals as est [lo, hi] where available.

Track 1 · option competition vs order



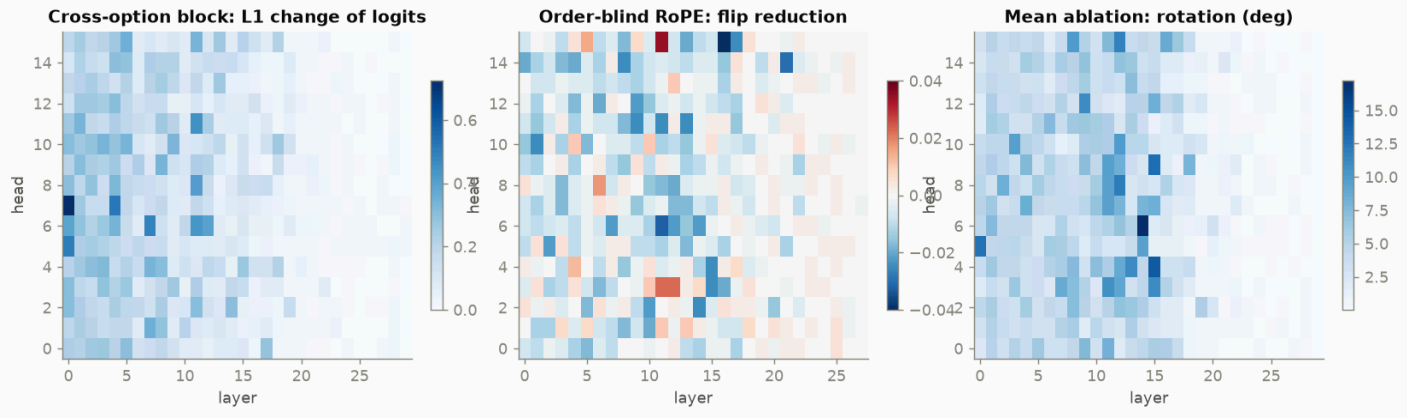
model	condition	accuracy	reversal flips	|M| centered
laya	sequential	0.456	0.289	0.375
laya	masked_parallel_all	0.343	0.299	0.000
laya	shipped_parallel_positions	0.407	0.014	0.176
laya	shipped_parallel_positions_headmask	0.409	0.018	0.175
laya	options_blocked_seqpos	0.415	0.281	0.181
laya	band_masked_parallel_early_0_8	0.372	0.371	0.283
laya	band_masked_parallel_mid_9_17	0.423	0.258	0.239
laya	band_masked_parallel_late_18_27	0.448	0.302	0.348
laya	band_masked_parallel_head_28_29	0.455	0.290	0.371
laya	band_masked_parallel_enc_global	0.387	0.326	0.244
laya	band_masked_parallel_enc_local	0.402	0.288	0.232
laya	band_masked_parallel_enc_all	0.343	0.295	0.006
smolaya	sequential	0.548	0.177	0.194
smolaya	masked_parallel_all	0.476	0.182	0.000
smolaya	shipped_parallel_positions	0.542	0.010	0.069
smolaya	shipped_parallel_positions_headmask	0.549	0.004	0.069
smolaya	options_blocked_seqpos	0.538	0.162	0.073
smolaya	band_masked_parallel_early_0_5	0.502	0.188	0.155
smolaya	band_masked_parallel_mid_6_12	0.549	0.210	0.139
smolaya	band_masked_parallel_late_13_19	0.550	0.167	0.140

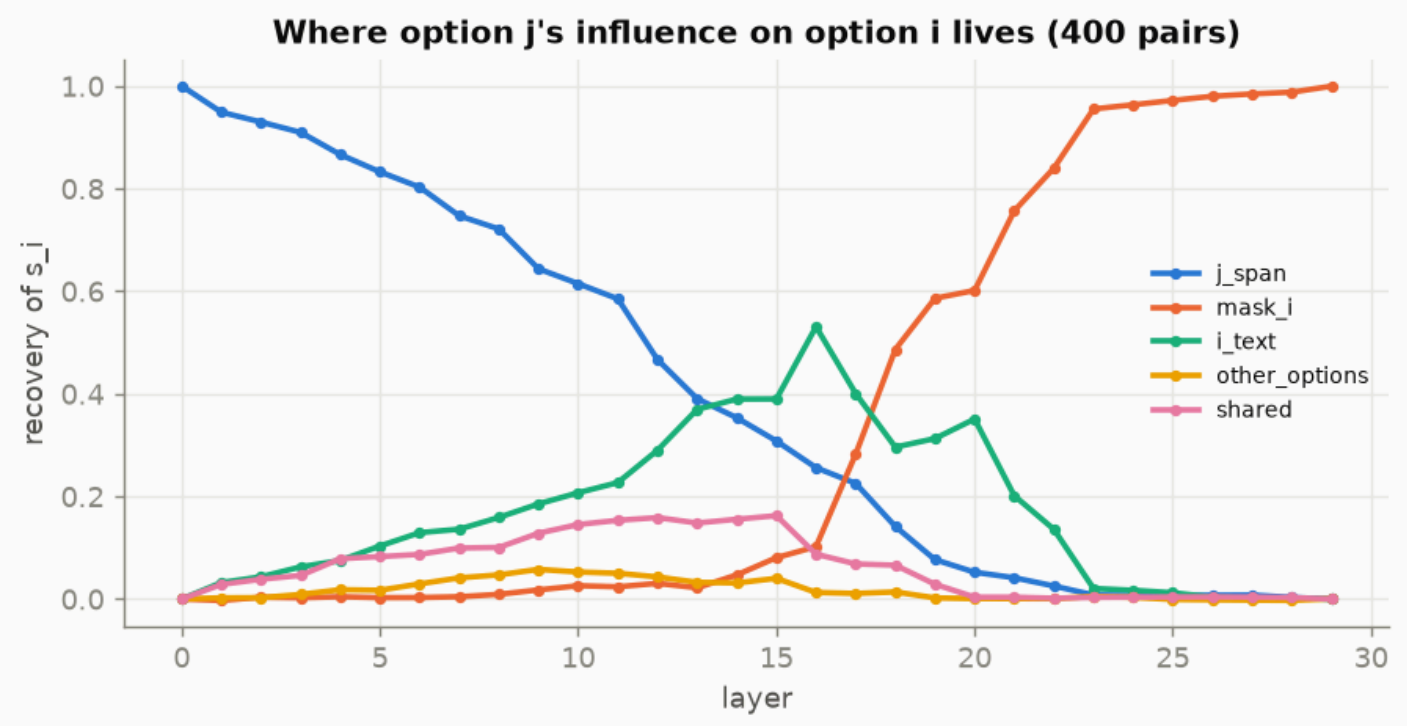
model	condition	accuracy	reversal flips	|M| centered
smolaya	band_masked_parallel_head_20_21	0.546	0.177	0.191
smolaya	band_masked_parallel_enc_global	0.521	0.192	0.113
smolaya	band_masked_parallel_enc_local	0.529	0.169	0.118
smolaya	band_masked_parallel_enc_all	0.476	0.177	0.005

Baselines (MC: ARC, CSQA, QASC)

model	split	accuracy	reversal flips	ECE raw	ECE tempered
laya	main	0.456 [0.430, 0.484]	0.289 [0.266, 0.314]	0.192	0.074
laya	heldout	0.430 [0.392, 0.469]	0.303 [0.265, 0.343]	0.233	0.100
laya	order map	slot R^2=0.002	0.555 [0.529, 0.580]	any-order flips →	neighbour r=-0.001
smolaya	main	0.548 [0.523, 0.574]	0.177 [0.157, 0.196]	0.057	0.057
smolaya	heldout	0.575 [0.536, 0.613]	0.137 [0.111, 0.164]	0.082	0.082
smolaya	order map	slot R^2=0.014	0.332 [0.307, 0.357]	any-order flips →	neighbour r=0.017

Track 1.3–1.4 · heads, information flow, ablations

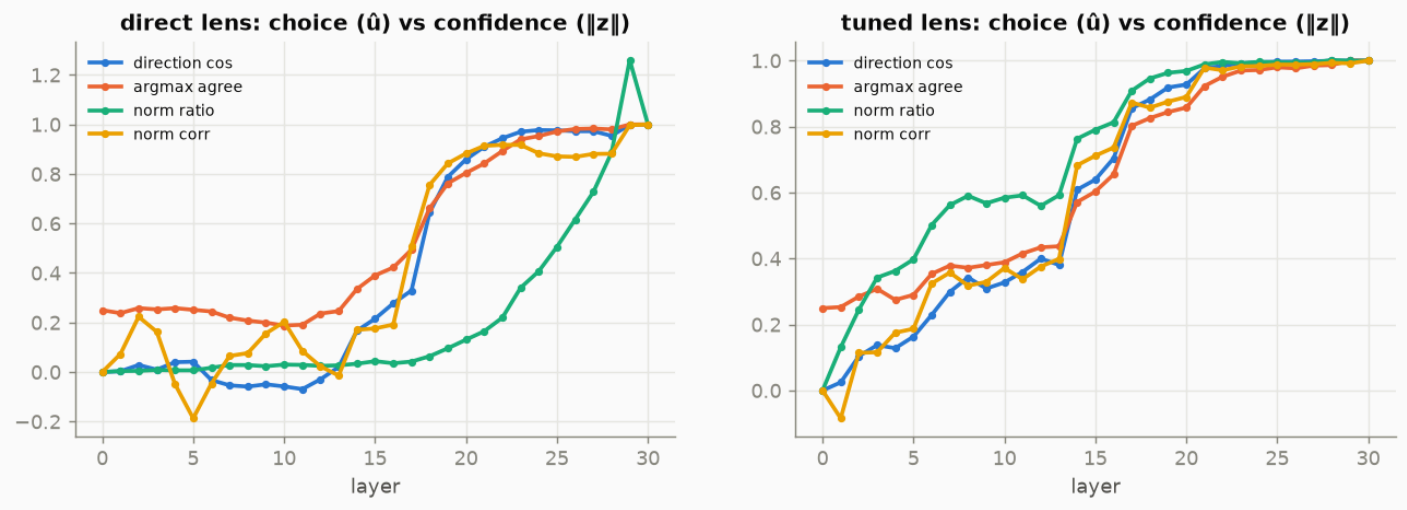


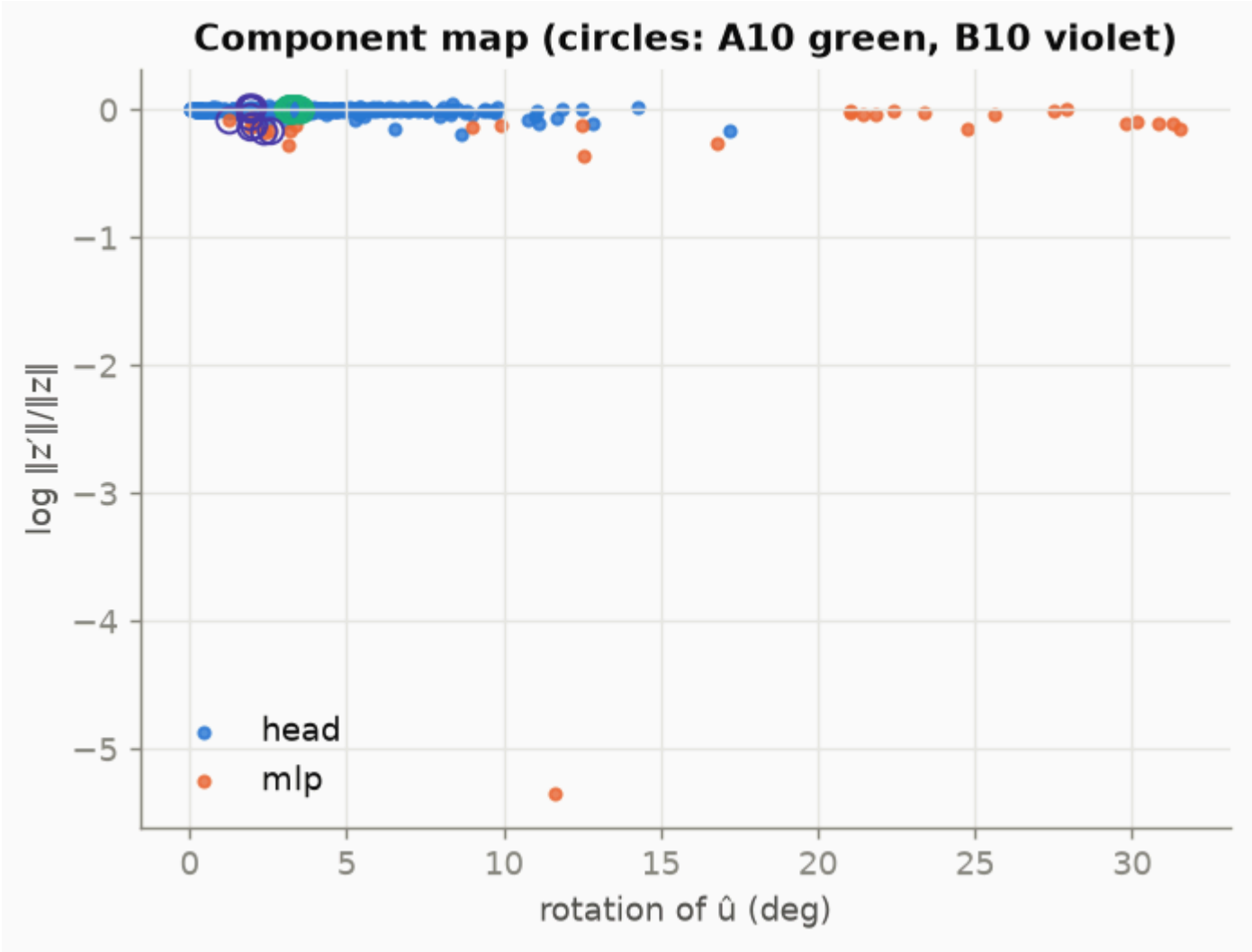


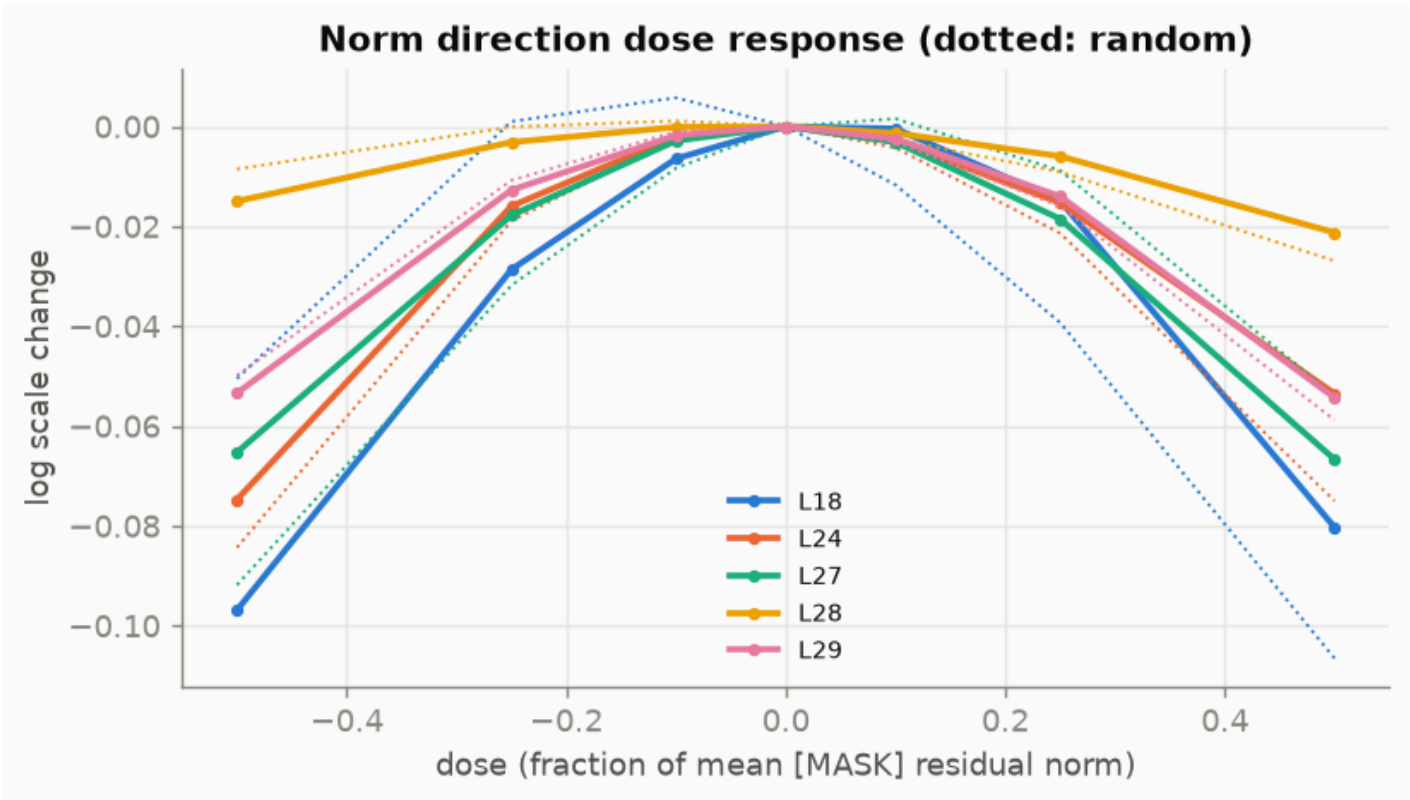
condition	accuracy	reversal_flip_rate	mean_abs_M_centered
base	0.456	0.289	0.371
rope_alt_all_enc_heads	0.229	0.083	0.000
cross_block_all_heads	0.415	0.281	0.180
comp5_cross_block	0.452	0.328	0.364
comp5_mean_ablate	0.448	0.274	0.358
comp5_resample	0.440	0.321	0.511
pos5_rope_alt	0.463	0.253	0.370
pos5_mean_ablate	0.422	0.292	0.329
rand5_s0_cross_block	0.454	0.299	0.369
rand5_s0_rope_alt	0.453	0.297	0.373
rand5_s0_mean_ablate	0.454	0.300	0.371
rand5_s1_cross_block	0.460	0.298	0.369
rand5_s1_rope_alt	0.458	0.291	0.370
rand5_s1_mean_ablate	0.465	0.297	0.374
rand5_s2_cross_block	0.453	0.294	0.372
rand5_s2_rope_alt	0.450	0.287	0.374
rand5_s2_mean_ablate	0.444	0.298	0.371
comp10_cross_block	0.430	0.338	0.358
comp10_mean_ablate	0.454	0.279	0.361
comp10_resample	0.433	0.334	0.584
pos10_rope_alt	0.460	0.267	0.365
pos10_mean_ablate	0.415	0.291	0.320
rand10_s0_cross_block	0.446	0.282	0.370
rand10_s0_rope_alt	0.440	0.297	0.369
rand10_s0_mean_ablate	0.438	0.297	0.345
rand10_s1_cross_block	0.452	0.293	0.370
rand10_s1_rope_alt	0.450	0.290	0.365
rand10_s1_mean_ablate	0.446	0.287	0.357
rand10_s2_cross_block	0.450	0.287	0.373
rand10_s2_rope_alt	0.453	0.296	0.371
rand10_s2_mean_ablate	0.449	0.300	0.373
comp20_cross_block	0.412	0.369	0.360
comp20_mean_ablate	0.443	0.261	0.344
comp20_resample	0.406	0.407	0.654
pos20_rope_alt	0.467	0.255	0.364
pos20_mean_ablate	0.412	0.288	0.313
rand20_s0_cross_block	0.448	0.288	0.349
rand20_s0_rope_alt	0.450	0.281	0.364
rand20_s0_mean_ablate	0.432	0.309	0.331

condition	accuracy	reversal_flip_rate	mean_abs_M_centered
rand20_s1_cross_block	0.446	0.285	0.367
rand20_s1_rope_alt	0.451	0.283	0.358
rand20_s1_mean_ablate	0.435	0.307	0.302
rand20_s2_cross_block	0.446	0.283	0.371
rand20_s2_rope_alt	0.453	0.281	0.368
rand20_s2_mean_ablate	0.444	0.278	0.334
heldout_base	0.430	0.303	0.389
heldout_comp10_cross_block	0.436	0.315	0.369
heldout_pos10_rope_alt	0.442	0.265	0.383
heldout_comp10_mean_ablate	0.419	0.270	0.364
heldout_pos10_mean_ablate	0.437	0.276	0.334

Track 2 · choice vs confidence





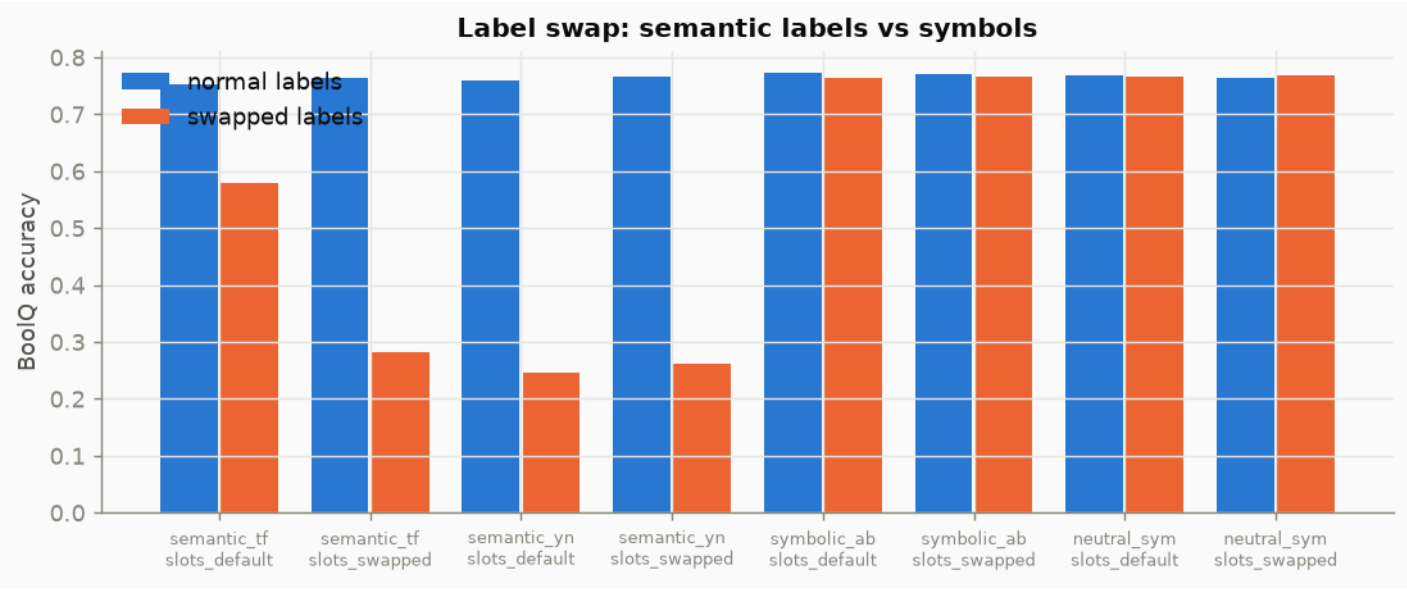


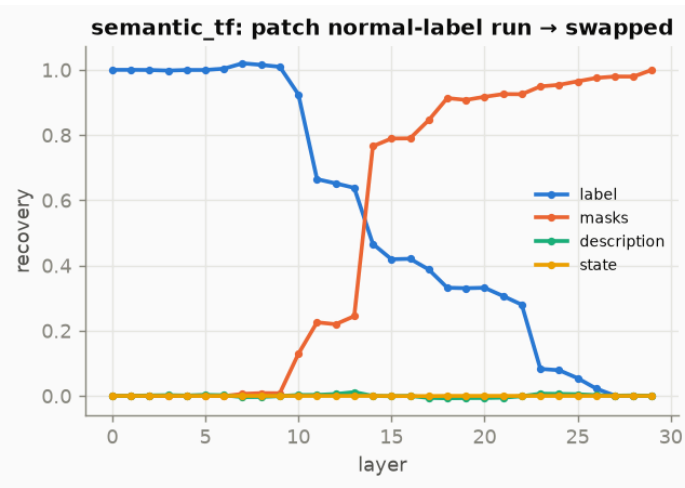
set	rotation°	|log scale|	log scale	argmax kept	accuracy	AUROC margin	ECE raw
A10_heldout	10.114 [9.275, 10.932]	0.120 [0.109, 0.130]	0.002 [-0.012, 0.016]	0.891	0.432	0.659	0.229

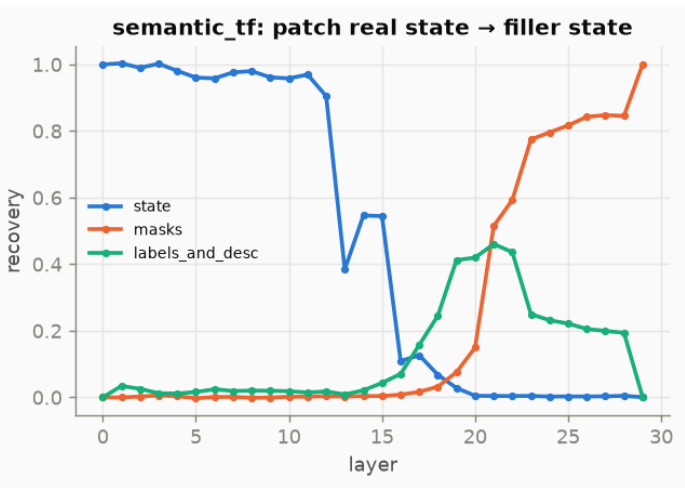
set	rotation°	|log scale|	log scale	argmax kept	accuracy	AUROC margin	ECE raw
A10_main	10.526 [9.940, 11.239]	0.126 [0.119, 0.133]	-0.002 [-0.014, 0.008]	0.889	0.462	0.628	0.188
A5_heldout	7.398 [6.843, 7.930]	0.102 [0.092, 0.114]	-0.011 [-0.026, 0.002]	0.916	0.434	0.653	0.226
A5_main	7.897 [7.443, 8.438]	0.094 [0.088, 0.100]	-0.007 [-0.016, 0.000]	0.926	0.453	0.635	0.192
B10_heldout	8.623 [7.996, 9.288]	1.234 [1.218, 1.250]	-1.234 [-1.250, -1.218]	0.930	0.429	0.665	0.028
B10_main	8.810 [8.345, 9.331]	1.244 [1.233, 1.256]	-1.244 [-1.256, -1.233]	0.919	0.459	0.635	0.061
B5_heldout	7.243 [6.741, 7.706]	1.236 [1.224, 1.247]	-1.236 [-1.247, -1.224]	0.938	0.437	0.653	0.042
B5_main	7.287 [6.871, 7.755]	1.245 [1.237, 1.255]	-1.245 [-1.255, -1.237]	0.928	0.453	0.646	0.058
rand10_s0_main	12.108 [11.488, 12.757]	0.144 [0.135, 0.153]	0.014 [0.003, 0.025]	0.895	0.453	0.642	0.200
rand10_s1_main	28.134 [26.786, 29.496]	0.288 [0.274, 0.303]	-0.010 [-0.029, 0.014]	0.732	0.416	0.651	0.230
rand10_s2_main	13.866 [13.191, 14.604]	0.168 [0.159, 0.178]	-0.067 [-0.080, -0.054]	0.861	0.449	0.645	0.186
rand5_s0_main	12.682 [12.095, 13.358]	0.238 [0.230, 0.248]	-0.199 [-0.211, -0.189]	0.873	0.443	0.628	0.152
rand5_s1_main	22.379 [21.233, 23.438]	0.251 [0.238, 0.262]	-0.091 [-0.109, -0.073]	0.777	0.448	0.622	0.174
rand5_s2_main	10.245 [9.722, 10.799]	0.124 [0.116, 0.131]	0.002 [-0.009, 0.012]	0.902	0.448	0.646	0.202

Overconfidence: global T=2.61; T by ||z|| quintile [1.78, 2.08, 2.3, 2.38, 2.97]; by source {'ARC-Challenge': 4.81, 'ARC-Easy': 2.0, 'CSQA': 2.48, 'QASC': 2.52}; ECE after global T 0.047.

Track 3 · label-following (BoolQ)

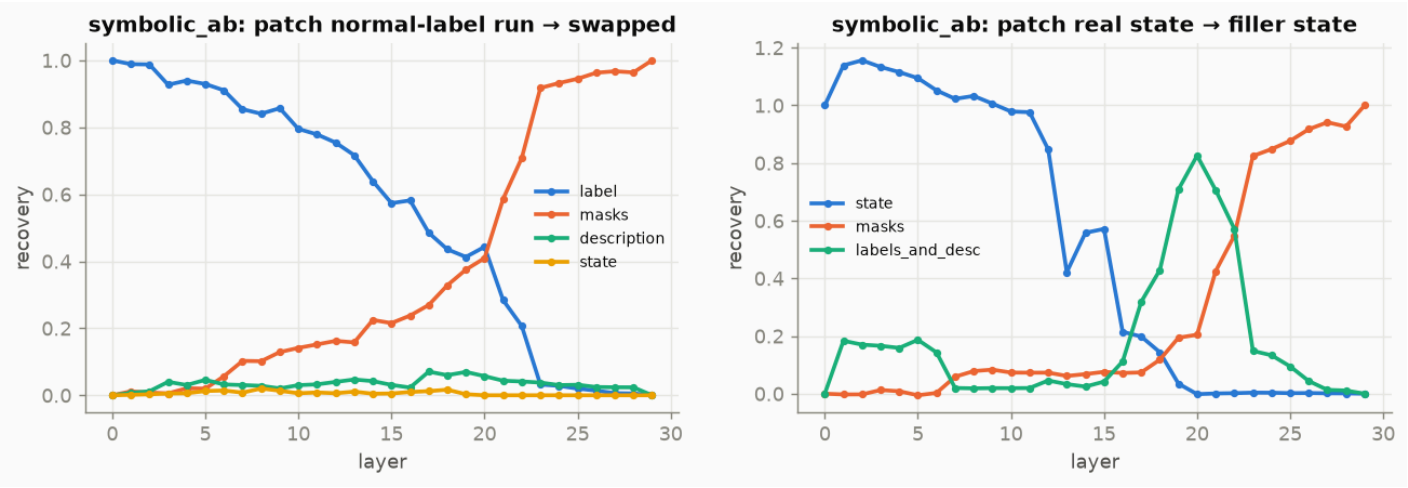






Edge knockouts (semantic_tf)

knockout	acc_normal	acc_swapped	label_following_rate	swap_gap
base	0.753	0.579	0.393	0.174
mask_to_labels@local	0.759	0.759	0.041	0.000
mask_to_labels@global	0.750	0.583	0.373	0.167
mask_to_labels@head	0.753	0.577	0.397	0.176
mask_to_labels@all	0.763	0.759	0.017	0.004
options_to_labels@local	0.757	0.749	0.043	0.009
options_to_labels@global	0.751	0.556	0.420	0.196
options_to_labels@head	0.753	0.577	0.397	0.176
options_to_labels@all	0.759	0.767	0.002	-0.009
mask_to_state@local	0.751	0.660	0.259	0.091
mask_to_state@global	0.754	0.659	0.248	0.096
mask_to_state@head	0.753	0.584	0.389	0.169
mask_to_state@all	0.751	0.689	0.194	0.063
options_to_state@local	0.751	0.664	0.251	0.087
options_to_state@global	0.751	0.680	0.215	0.071
options_to_state@head	0.753	0.583	0.389	0.170
options_to_state@all	0.751	0.699	0.183	0.053
label_path_all	0.759	0.767	0.002	-0.009



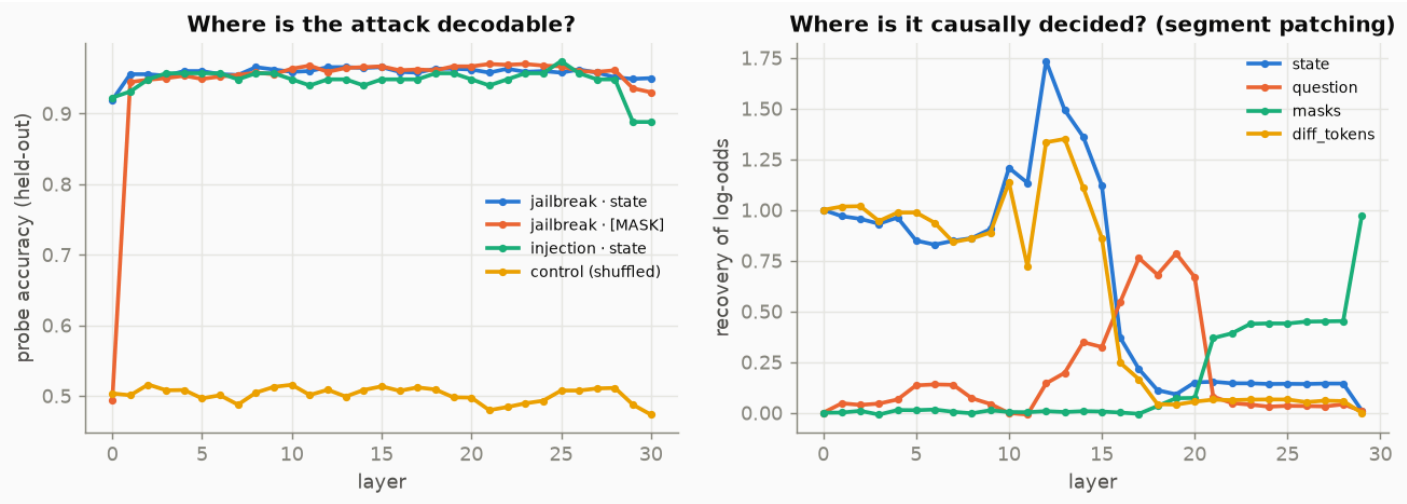
Edge knockouts (symbolic_ab)

knockout	acc_normal	acc_swapped	label_following_rate	swap_gap
base	0.773	0.764	0.033	0.009
mask_to_labels@local	0.763	0.764	0.007	-0.001
mask_to_labels@global	0.771	0.769	0.019	0.003
mask_to_labels@head	0.773	0.764	0.033	0.009
mask_to_labels@all	0.773	0.773	0.009	0.000
options_to_labels@local	0.770	0.769	0.004	0.001

knockout	acc_normal	acc_swapped	label_following_rate	swap_gap
options_to_labels@global	0.774	0.773	0.018	0.001
options_to_labels@head	0.773	0.764	0.033	0.009
options_to_labels@all	0.767	0.764	0.004	0.003
mask_to_state@local	0.767	0.763	0.028	0.004
mask_to_state@global	0.761	0.761	0.041	0.000
mask_to_state@head	0.774	0.761	0.035	0.013
mask_to_state@all	0.757	0.766	0.032	-0.009
options_to_state@local	0.773	0.766	0.037	0.007
options_to_state@global	0.760	0.760	0.051	0.000
options_to_state@head	0.774	0.761	0.035	0.013
options_to_state@all	0.770	0.753	0.067	0.017
label_path_all	0.767	0.764	0.004	0.003

pad tokens	semantic_tf follow	symbolic_ab follow	semantic_tf acc	symbolic_ab acc
0	0.385	0.033	0.760	0.767
64	0.567	0.053	0.750	0.752
128	0.745	0.033	0.745	0.755
256	0.771	0.039	0.752	0.765

Track 4 · jailbreak and injection



α	benign → jailbreak frac	urgency frac (specificity)	MC accuracy
0	0.213	0.000	0.420
1	0.160	0.000	0.312
2	0.003	0.000	0.247
4	0.000	0.000	0.253
8	0.000	0.000	0.253

ablation	auroc	accuracy	decision_flips	mean_abs_logodds_ratio
base	0.943	0.877	0.000	1.000
comp20_cross_block	0.944	0.877	0.007	1.000

ablation	auroc	accuracy	decision_flips	mean_abs_logodds_ratio
comp20_mean	0.908	0.836	0.079	1.002
pos20_mean	0.890	0.826	0.078	0.962
rand20_s0_mean	0.895	0.813	0.081	0.937
rand20_s1_mean	0.924	0.849	0.068	0.912
rand20_s2_mean	0.919	0.834	0.058	1.078
A10_mean	0.937	0.887	0.026	0.963
B10_mean	0.860	0.828	0.056	0.361

Baselines: natural AUROC 0.947, injection AUROC 0.867; pairs kept 1016/1700; cos(jailbreak, injection direction) 0.710.

Track 4.3 · English direction on Hindi (laya-multilingual)

Flagged: English jailbreak 0.81 vs control 0.20; Hindi jailbreak 0.52 vs control 0.09. English direction at L16: Hindi projection AUROC 0.561; projecting it out of Hindi jailbreaks: 0.52 → 0.09 flagged; random direction at α=8: 0.04.

α (σ)	Hindi controls flagged	mean log-odds
0	0.085	-5.657
1	0.085	-5.772
2	0.085	-5.788
4	0.105	-5.339
8	0.125	-5.138

Follow-ups: fp32, held-out, greedy sets, steering, LLM data

Greedy C2 sets

set	components	main abs_log_scale	main argmax_unchanged	main ece_raw	heldout abs_log_scale	heldout argmax_unchanged	heldout ece_raw
B	mlp L22, mlp L27, mlp L26	0.733	0.960	0.078	0.739	0.966	0.080
A	head L10 h2, head L6 h6, head L3 h14, head L6 h1, head L18 h7, head L16 h6	0.091	0.903	0.203	0.088	0.941	0.227

Steering sweep

Base: {'benign_flagged': 0.23, 'jb_flagged': 0.985, 'mc_acc': 0.42}. Best specific add: {'layer': 20, 'segment': 'masks', 'alpha_gap_units': 1.0, 'gap_over_resid_norm': 0.16951527740109223, 'benign_flagged': 0.61, 'jb_flagged_after_neg': 0.93, 'urgency_flagged': 0.03, 'mc_acc': 0.42}. Best specific suppress: {'layer': 12, 'segment': 'state', 'alpha_gap_units': 2.0, 'gap_over_resid_norm': 0.45843564778091994, 'benign_flagged': 0.38, 'jb_flagged_after_neg': 0.72, 'urgency_flagged': 0.0, 'mc_acc': 0.39666666666666667}.

Held-out Track 3

Knockouts: base {'acc_normal': 0.77, 'acc_swapped': 0.5466666666666666, 'label_following_rate': 0.4199134199134199, 'swap_gap': 0.22333333333333338}; local mask→label {'acc_normal': 0.7766666666666666, 'acc_swapped': 0.77, 'label_following_rate': 0.030042918454935622, 'swap_gap': 0.006666666666666599}.

fp32 confirmation

(Raw fp32 output omitted; the fp32 numbers appear in the main text.)

LLM-generated data ({'bulk': 'gpt-5.6-luna', 'careful': 'gpt-5.6-terra'})

Counterfactuals: {'n': 145, 'orig_flagged': 1.0, 'edited_flagged': 0.8206896551724138, 'mean_logodds_drop': 7.707245826721191}.

labels	acc normal	acc swapped	label-following
semantic_tf/slots_default	0.785	0.455	0.598
semantic_tf/slots_swapped	0.800	0.265	0.895
semantic_yn/slots_default	0.788	0.230	0.963
semantic_yn/slots_swapped	0.800	0.243	0.930
symbolic_ab/slots_default	0.803	0.822	0.033
symbolic_ab/slots_swapped	0.806	0.810	0.022
neutral_sym/slots_default	0.803	0.811	0.018
neutral_sym/slots_swapped	0.813	0.826	0.007

Knockout (local mask→label): {'acc_normal': 0.8114874815905744, 'acc_swapped': 0.812960235640648, 'label_following_rate': 0.041742286751361164, 'swap_gap': -0.0014727540500736325}; base {'acc_normal': 0.7849779086892489, 'acc_swapped': 0.45508100147275404, 'label_following_rate': 0.5984990619136961, 'swap_gap': 0.32989690721649484}. Usage {'prompt_tokens': 159516, 'completion_tokens': 186334, 'calls': 1293}.

Stretch · same init, same data: soft-CE vs RLCD

model	accuracy	reversal flips	flips, shared positions	flips, options blocked	ECE raw	|M| centered	greedy-B MLPs	label-following yes/no
ctrl_soft_ce	0.713 [0.690, 0.736]	0.133 [0.115, 0.152]	0.008	0.152	0.125	0.433 [0.424, 0.441]	[28, 22, 23]	0.052
ctrl_rlcd	0.690 [0.665, 0.713]	0.149 [0.131, 0.167]	0.007	0.154	0.100	0.345 [0.338, 0.352]	[28, 22, 23, 29]	0.003

SAE · [MASK] features

L21: FVE 1.000, recon keeps 0.953 of answers, top-25 choice∩confidence overlap 11; L28: FVE 0.974, recon keeps 0.993 of answers, top-25 choice∩confidence overlap 15

layer	ablation	abs_log_scale	log_scale	rotation_deg	argmax_kept
L21	ablate_conf10	0.276	-0.273	6.371	0.953
L21	ablate_choice10	0.390	-0.388	7.331	0.923
L21	ablate_random10	0.001	-0.001	0.085	1.000
L28	ablate_conf10	0.378	-0.269	13.540	0.941
L28	ablate_choice10	1.831	-1.831	27.405	0.745
L28	ablate_random10	0.001	-0.000	0.043	0.998